Dataset Description: Heart Disease Analysis

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1. Dataset Overview

Name: Heart Disease Dataset.

Source: Kaggle [https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset].

File Format: CSV.

Rows and Columns: 1025, 14.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
52	1	0	125	212	0	1	168	0	1	2	2	3	0
53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
61	1	0	148	203	0	1	161	0	0	2	1	3	0
62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
58	0	0	100	248	0	0	122	0	1	1	0	2	1
58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
55	1	0	160	289	0	0	145	1	0.8	1	1	3	0
46	1	0	120	249	0	0	144	0	0.8	2	0	3	0
54	1	0	122	286	0	0	116	1	3.2	1	2	2	0
71	0	0	112	149	0	1	125	0	1.6	1	0	2	1
43	0	0	132	341	1	0	136	1	3	1	0	3	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	1	0	140	298	0	1	122	1	4.2	1	3	3	0
52	1	0	128	204	1	1	156	1	1	1	0	0	0
34	0	1	118	210	0	1	192	0	0.7	2	0	2	1
51	0	2	140	308	0	0	142	0	1.5	2	1	2	1
54	1	0	124	266	0	0	109	1	2.2	1	1	3	0
50	0	1	120	244	0	1	162	0	1.1	2	0	2	1
58	1	2	140	211	1	0	165	0	0	2	0	2	1
60	1	2	140	185	0	0	155	0	3	1	0	2	0
67	0	0	106	223	0	1	142	0	0.3	2	2	2	1
45	1	0	104	208	0	0	148	1	3	1	0	2	1
63	0	2	135	252	0	0	172	0	0	2	0	2	1
42	0	2	120	209	0	1	173	0	0	1	0	2	1

2. Features Description

- Age: Patient's age in years.
- Sex: Patient's biological gender (1 for male, 0 for female)
- Chest Pain Type: Type of chest pain (classic angina, atypical angina, non-anginal, no pain).
- Resting Blood Pressure: Patient's resting systolic pressure (mm Hg).
- Cholesterol: Blood cholesterol levels (mg/dL).
- Fasting blood sugar: testing if the patient's blood sugar is >120 mg/dL (1 for True; 0 for False).
- Resting Electrocardiographic Results: ECG results of the heart's electrical function (resting).
- Maximum Heart Rate Achieved: Max HR reached during exercise testing.
- Exercise-Induced Angina: Chest discomfort occurred during exercise testing (1 for yes, 0 for no)

- ST Depression (Oldpeak): This is a type of ST elevation (or depression) measured during exercise on the electrocardiogram when compared to what was recorded when the patient is resting.
- Peak Exercise ST Segment Slope: This describes the slope of the ST segment will be recorded on the electrocardiogram by running the patient at maximum exertion (exercise) in three different slopes (example: upward slope, flat slope, downward slope, etc.)
- Number of Major Coronary Arteries: The number of major coronary arteries (0-3) visualized by fluoroscopy during the patient's exercise test.
- Thallasimia Status: The status of thallasimia is either normal, fixed defect, and reversible defect.
- Target (Diagnosis): This is the output label that indicates the presence of heart disease (1 = heart disease detected; 0 = no heart disease detected).

3. Purpose of Using This Dataset

- Clinical Relevance: Heart diseases kill more people than just anything else. Digging into this dataset lets us see firsthand how AI can help catch these problems early.
- Feature Diversity: The dataset contains a balanced mix of categorical (e.g., chest pain type) and numerical (e.g., maximum heart rate) features, making it an ideal candidate for practicing complex Data Visualization and Correlation Analysis.
- Real-World Application: This dataset is a benchmark in the machine learning community. Working with it helps in building foundational skills for health-tech AI applications.
- Data Integrity Challenges: I chose this version of the dataset specifically because it requires careful preprocessing (such as handling duplicates and verifying feature scales), which is a core skill in the Data Science lifecycle.